C3.6 Snoop address channel signaling

The following sections define the snoop address channel and signals.

C3.6.1 About the snoop address channel

The snoop address channel (AC channel) is necessary for a master that:

- holds cached copies of shared data
- supports DVM transactions.

The snoop address channel is an input channel to a cached master. The snoop address channel passes the snoop transactions of other components to a cached master so that the master can determine what actions it must take. The master can respond to the snoop transactions in different ways and its response determines what action the interconnect must take to complete the snoop process.

Supplementary information

Control information on the snoop address channel is a subset of the control information on the normal address channel. It provides sufficient information for the coherency operations, but does not provide unnecessary information. The snoop address channel does not provide information on:

- the burst type
- the burst length
- the transaction size
- the modifiable or shareable nature of the transaction
- the transaction ID.

Fundamentally, the snoop address channel provides the same transactions that are issued on the normal read and write address channels. However, there are a number of exceptions.

The following transactions are not presented on the snoop address channel:

- Non-snooping type transactions. These transactions are:
  - ReadNoSnoop
  - WriteNoSnoop
  - WriteBack
  - WriteClean
  - Evict.
- WriteUnique. Other cached masters see a transaction, such as CleanInvalid. This ensures that all other copies of the cache line are cleaned to main memory and removed before the write occurs.
- WriteLineUnique. Other cached masters see a transaction, such as MakeInvalid. This ensures that all other copies of the cache line are removed before the write occurs.
- MakeUnique. Typically, this transaction is converted to a MakeInvalid transaction.
- CleanUnique. Typically, this transaction is converted to a CleanInvalid transaction.

Some snoop operations can be fulfilled without snooping every cached master in the system. Therefore, the snoop address channel for a cached master might not provide every snoop transaction.

Snoop read transactions can be adapted by the interconnect so that when the required data is obtained, further snoops to other masters are not requested.

Transactions that are not required to be a full cache line length are converted to be a full cache line length. These transactions are:

- ReadOnce
- WriteUnique.

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C3.6.2 Snoop address channel signaling

Table C3-18 shows the signals on the snoop address channel.

Table C3-18 Snoop address channel signals

Signal	Source	Description
ACVALID	Interconnect	Snoop address valid. When HIGH, it indicates that the snoop address and control information is valid.
ACREADY	Master	Snoop address ready. When HIGH, it indicates that the snoop address and control information can be accepted in this cycle.
ACADDR[ac-1:0][a]	Interconnect	Snoop address. This signal indicates the address of a snoop transaction. The snoop address width must match the width of the read address and write address bus.
ACSNOOP[3:0]	Interconnect	Snoop transaction type. This signal indicates a subset of the transaction types observed on the read and write address channels. See Table C3-19.
ACPROT[2:0]	Interconnect	Snoop protection type. This signal indicates the privilege and security level of the snoop transaction.

a. ac is the width of the snoop address bus.

The standard AXI **VALID**/**READY** handshake rules apply. See *Handshake process* on page A3-37.

When the **ACVALID** signal is asserted the snoop address and control signals on **ACADDR**, **ACPROT**, and **ACSNOOP** must not change until **ACREADY** is asserted by the master. When **ACVALID** is asserted, it must remain asserted until **ACREADY** is asserted.

It is permitted to assert **ACREADY** before or in the same cycle as **ACVALID**. If **ACREADY** is asserted before **ACVALID** then **ACREADY** can be deasserted without **ACVALID** being asserted.

ACADDR must be aligned to the data transfer size, which is determined by the width of the snoop data bus in bytes.

ACPROT[1] indicates the Secure or Non-secure nature of the snoop transaction.

For coherency transactions, **ACPROT[1]** can be considered as defining two address spaces, a secure address space and a non-secure address space, and can be treated as an additional address bit. Any aliasing between the secure and non-secure address spaces must be handled correctly.

Hardware coherency does not manage coherency between secure and non-secure address spaces.

ACSNOOP indicates the snoop transaction type. Not all transaction types, observed on the read address channel or write address channel can be observed on the snoop address channel. Table C3-19 shows the **ACSNOOP** encodings for the transactions that can be observed on the snoop address channel. All unused encodings are Reserved.

Table C3-19 ACSNOOP encodings

ACSNOOP[3:0]	Transaction
0b0000	ReadOnce
0b0001	ReadShared
0b0010	ReadClean
0b0011	ReadNotSharedDirty
0b0111	ReadUnique
0b1000	CleanShared
0b1001	CleanInvalid

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Table C3-19 ACSNOOP encodings (continued)

ACSNOOP[3:0]	Transaction
0b1101	MakeInvalid
0b1110	DVM Complete
0b1111	DVM Message

A snoop transaction of burst length greater than one must be of burst type WRAP. A snoop transaction of burst length one must be of burst type INCR.

A snoop transaction must be a full cache line in length.

Note

If the original transaction that caused the snoop process was not a full cache line in length, then the interconnect must convert it to be a full cache line in length.

A snoop transaction must be the same width as the snoop data channel.

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